Rakan Eyad

Control Systems & GNC Engineer
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Summary

Mechatronic engineering graduate (Honours, University of Sydney) focused on control systems and GNC. Honours thesis coupled an A* global planner to a PPO local policy for planetary-rover navigation in NVIDIA Isaac Sim (78/100, Distinction). Built a two-axis vision tracking rig running one PID loop per axis against a Kalman velocity estimate, and spent 14 months at ALBON integrating and re-architecting a prototype control system.

Experience

System Engineer — ALBON Nov 2024 — Jan 2026
- Developed and integrated hardware, software, and mechanical components for a prototype control system: migrated the architecture from a microcontroller to a Raspberry Pi for real-time data processing, integrated sensor arrays, and engineered the system's electrical junction box to cut power consumption and keep the hardware reliable.

Engineering Intern — Escape The Room Sep 2021 — Nov 2021
- Designed mechanical puzzles and implemented engineering concepts in real-world, client-facing products.

Testing Technician Intern — CloudAk May 2021 — Aug 2021
- Supported server system testing and maintenance in data-centre environments.

Featured Projects

Autonomous Target Tracking System — Personal build · hardware
- Designed, built, wired, and tuned a two-axis vision tracking rig end to end: a closed-loop pointing system holding a moving hand or drone inside a bounding box at the centre of frame, with no off-the-shelf motion controller.
- Ran one PID loop per motor against a Kalman filter estimating target velocity, so the mount keeps leading the target through detection dropouts instead of losing lock.

Planetary Rover RL Navigation — Thesis · Isaac Sim
- Honours thesis (78/100, Distinction): a hybrid guidance-and-control architecture coupling an A* global planner to a PPO local policy through a pure-pursuit waypoint interface, trained in NVIDIA Isaac Sim across 64 parallel rovers on Martian terrain.
- Hybrid held 99.5% success at the 10 m training distance against a standalone learned policy's 79.5%, and 90% at 50 m where the standalone collapsed to 2%.
- Authored the A* command term, terrain importer, skrl training configs, reward and termination tuning, and the evaluation pipeline; built on the open-source RLRoverLab codebase.

Assistive Robot (FPGA) — University build · Verilog
- Implemented closed-loop control directly in FPGA fabric on an Altera DE2-115: a camera line-follower and object tracker driving wheel commands, with ultrasonic and IR safety interlocks arbitrated by a four-state FSM (team of 5).
- Closed timing by pipelining the critical path, turning a 32.763 ns violation into 6.46 ns of positive slack; all tasks demonstrated on physical hardware.

Conjure — Open source · Python
- Built a real-time vision pipeline classifying 9 hand/face gestures from MediaPipe landmarks, stabilised by a two-stage One Euro + EMA filter with temporal hysteresis to hold a steady output at 60 FPS.
- Profiled and optimized the render path from 162 ms to 2.8 ms per frame (58x speed-up) through allocation discipline.

Skills

Control & signals: Advanced control systems, Signal processing & filtering, State estimation, GNC fundamentals, MATLAB, Radar & antenna coursework
Robotics & simulation: ROS2, Isaac Sim / Isaac Lab, State estimation & path planning, HIL simulation, CAD (Fusion/SW), Mechanism design
Firmware & electronics: C / C++ · Arduino, FPGA · Verilog / SystemVerilog, Raspberry Pi · microcontrollers, Sensor integration & calibration, Power wiring & junction boxes, Assembly

Education

B.E. (Honours), Mechatronic Engineering
University of Sydney · 2022 — 2026
Graduated with First Class Honours